Problem 2b: Answer Format
(Exactly What to Write)

ECO 310 - The Format You Need

February 23, 2026

1 What Problem 2b Asks

Question: Write out the optimality conditions, *if you guess that the utility constraint is binding, nonnegativity on good 1 is binding, and nonnegativity on good 2 is non-binding.*

Translation: Given this specific guess, what do the equations look like?

2 The Guess in Math Terms

The guess tells you:

- **Utility constraint is binding:** $x_1 + x_2 = u^*$ (exactly, not $>$)
- **Nonnegativity on good 1 is binding:** $x_1 = 0$ (exactly)
- **Nonnegativity on good 2 is non-binding:** $x_2 > 0$ (strictly positive)

What this means for the λ values:

- Since $x_1 = 0$ (constraint binds), $\lambda_1 \geq 0$ (can be positive)
- Since $x_2 > 0$ (constraint doesn't bind), $\lambda_2 = 0$ (complementary slackness!)
- Since utility constraint binds, $\lambda_3 \geq 0$ (can be positive)

3 Format of Your Answer

Here's EXACTLY what you should write for Problem 2b:

Problem 2b Answer

Given the guess:

- Utility constraint is binding: $x_1 + x_2 = u^*$
- Nonnegativity on good 1 is binding: $x_1 = 0$
- Nonnegativity on good 2 is non-binding: $x_2 > 0$

The optimality conditions simplify to:
1. From $\frac{\partial \mathcal{L}}{\partial x_1}$ (since $x_1 = 0$):

$$-p_1 + \lambda_1 + \lambda_3 \leq 0 \tag{1}$$

2. From $\frac{\partial \mathcal{L}}{\partial x_2}$ (since $x_2 > 0$):

$$-p_2 + \lambda_2 + \lambda_3 = 0 \tag{2}$$

Since $\lambda_2 = 0$ (complementary slackness with $x_2 > 0$):

$$-p_2 + \lambda_3 = 0 \quad \Rightarrow \quad \lambda_3 = p_2 \tag{3}$$

3. From $\frac{\partial \mathcal{L}}{\partial \lambda_3}$ (utility constraint):

$$x_1 + x_2 = u^* \tag{4}$$

Since $x_1 = 0$:

$$x_2 = u^* \tag{5}$$

4. Solution from these conditions:

$$x_1 = 0 \tag{6}$$
$$x_2 = u^* \tag{7}$$
$$\lambda_1 = p_1 - p_2 \quad \text{(from equation 1, substituting } \lambda_3 = p_2\text{)} \tag{8}$$
$$\lambda_2 = 0 \tag{9}$$
$$\lambda_3 = p_2 \tag{10}$$

5. For this to be valid, we need $\lambda_1 \geq 0$:

$$\lambda_1 = p_1 - p_2 \geq 0 \quad \Rightarrow \quad p_1 \geq p_2 \tag{11}$$

This solution is valid when good 1 is at least as expensive as good 2.

4 Step-by-Step Logic

Here's how you get to that answer:

4.1 Step 1: Identify What You Know

From the guess:

- $x_1 = 0$ (given)

- $x_2 > 0$ (given)
- $x_1 + x_2 = u^*$ (given)

From complementary slackness:

- Since $x_2 > 0$, we must have $\lambda_2 = 0$
- Since $x_1 = 0$, we can have $\lambda_1 > 0$

4.2 Step 2: Write the General Optimality Conditions

From your Lagrangian $\mathcal{L} = -p_1x_1 - p_2x_2 + \lambda_1x_1 + \lambda_2x_2 + \lambda_3(x_1 + x_2 - u^*)$

General conditions are:

$$\frac{\partial \mathcal{L}}{\partial x_1} = -p_1 + \lambda_1 + \lambda_3 \leq 0, \quad x_1 \geq 0, \quad x_1(-p_1 + \lambda_1 + \lambda_3) = 0 \tag{12}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = -p_2 + \lambda_2 + \lambda_3 \leq 0, \quad x_2 \geq 0, \quad x_2(-p_2 + \lambda_2 + \lambda_3) = 0 \tag{13}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_3} = x_1 + x_2 - u^* \geq 0, \quad \lambda_3 \geq 0, \quad \lambda_3(x_1 + x_2 - u^*) = 0 \tag{14}$$

4.3 Step 3: Apply Your Guess to Simplify

For x_1 condition:

- Since $x_1 = 0$, the complementary slackness $x_1(-p_1 + \lambda_1 + \lambda_3) = 0$ is satisfied automatically
- We just need: $-p_1 + \lambda_1 + \lambda_3 \leq 0$

For x_2 condition:

- Since $x_2 > 0$, complementary slackness requires: $-p_2 + \lambda_2 + \lambda_3 = 0$ (exactly zero!)
- Since we know $\lambda_2 = 0$: $-p_2 + \lambda_3 = 0$, so $\lambda_3 = p_2$

For utility constraint:

- Since constraint binds: $x_1 + x_2 = u^*$
- Since $x_1 = 0$: $x_2 = u^*$

4.4 Step 4: Solve for Everything

Now substitute $\lambda_3 = p_2$ into the x_1 condition:

$$-p_1 + \lambda_1 + p_2 \leq 0 \quad \Rightarrow \quad \lambda_1 \leq p_1 - p_2 \tag{15}$$

For $\lambda_1 \geq 0$ (required), we need $p_1 - p_2 \geq 0$, or $p_1 \geq p_2$.

Actually, at the optimal solution, we typically have the inequality binding, so:

$$\lambda_1 = p_1 - p_2 \tag{16}$$

Wait, that gives $\lambda_1 \geq 0$ when $p_1 \geq p_2$. But this guess assumes we buy only good 2 ($x_1 = 0$, $x_2 = u^*$), which happens when good 2 is cheaper or equal price!

Let me recalculate...

If $\lambda_3 = p_2$ and $-p_1 + \lambda_1 + \lambda_3 \leq 0$:

$$\lambda_1 \leq p_1 - p_2 \tag{17}$$

Setting to equality (binding): $\lambda_1 = p_1 - p_2$

For this to be valid ($\lambda_1 \geq 0$): $p_1 \geq p_2$

Hmm, but intuitively if $x_1 = 0$ and $x_2 = u^*$, we're buying only good 2, which should happen when good 2 is cheaper ($p_2 < p_1$).

Actually, checking the inequality: $\lambda_1 = p_1 - p_2$. If $p_1 > p_2$, then $\lambda_1 > 0$, which is allowed.

Conclusion: The guess $x_1 = 0$, $x_2 = u^*$ is valid when $p_1 \geq p_2$ (good 1 is at least as expensive).

5 What to Actually Write on Your Homework

Just write what's in the yellow box above! That's the format your professor expects.

The key parts are:

1. State the guess clearly
2. Write simplified optimality conditions given that guess
3. Solve the system of equations
4. State the final solution: values of $x_1, x_2, \lambda_1, \lambda_2, \lambda_3$
5. Check validity (when does $\lambda_1 \geq 0$?)

6 Common Confusion

Q: Why does $x_2 > 0$ mean $\lambda_2 = 0$?

A: Complementary slackness! The condition is $\lambda_2 \cdot x_2 = 0$. If $x_2 > 0$, then we must have $\lambda_2 = 0$ for the product to equal zero.

Q: Why does $x_1 = 0$ mean λ_1 can be positive?

A: Complementary slackness again: $\lambda_1 \cdot x_1 = 0$. If $x_1 = 0$, then λ_1 can be anything (including positive) and the product is still zero.

Q: Where does $\lambda_3 = p_2$ come from?

A: From the FOC for x_2: $-p_2 + \lambda_2 + \lambda_3 = 0$. Since $\lambda_2 = 0$, we get $\lambda_3 = p_2$.